

Student Codebook: AI Use, Mental Health, and Wellbeing

AI Student Wellbeing Dataset (N = 400)

Study Context and Measurement Overview

This codebook describes the variables in the AI Student Wellbeing dataset. The dataset is organized as one row per college student and includes measures of academic AI use, perceived benefits and concerns about AI, critical thinking and self-verification practices, academic context, time and financial pressures, and student mental health and wellbeing.

The study context is the rapid integration of artificial intelligence into higher education. Recent reviews describe AI in higher education as a broad category of tools used for assessment and evaluation, prediction, AI assistance, intelligent tutoring, and managing student learning. AI tools can support students by adapting instruction, providing feedback, and organizing learning, but the same tools also raise concerns about accuracy, privacy, academic integrity, over-reliance, and the possible weakening of independent thinking.

Most attitudinal and experience items use a 1 to 5 response scale. For these variables, higher values mean more of the named construct. For example, higher values on stress items mean more stress, higher values on social support mean more perceived support, and higher values on AI benefit items mean stronger perceived benefits. Composite indexes summarize several related survey items into a single score so that the broader construct can be examined alongside the individual items.

Literature Basis for the Measures

The AI-related constructs draw on two sources. Crompton and Burke's systematic review of AI in higher education describes a rapidly expanding field and identifies common higher education uses of AI, including AI assistants, intelligent tutoring systems, assessment/evaluation, prediction, and learning management. Vieriu and Petrea's study of student academic development emphasizes both benefits - such as personalized learning, improved engagement, time savings, feedback, and academic confidence - and risks, including incorrect answers, over-dependence, reduced critical thinking, privacy concerns, and academic-integrity ambiguity.

The mental health and wellbeing constructs are not clinical diagnoses. They are self-report indicators of students' recent school-related experiences, such as feeling overwhelmed, anxious about school or the future, emotionally drained, socially connected, or optimistic. These variables are intended to support questions about student wellbeing and study validity rather than to classify individual students medically.

Scale Direction and Missing Context

Unless otherwise noted, 1 to 5 survey items are coded so that higher numbers mean more of the experience named in the variable. For example, higher values on benefit_efficiency mean stronger agreement that AI improves efficiency, while higher values on concern_accuracy mean greater concern about inaccurate information.

The codebook defines what each variable is intended to capture. It does not tell you which variables should be placed in any specific statistical role. In your analysis, you should justify your own choices based on the research question, measurement validity, and plausible alternative explanations.

Key Variables and Construct Context

The sections below provide richer context for the main constructs in the dataset. Variable names listed under each heading indicate the dataset columns connected to that construct.

Academic AI Use and Exposure

Variables: ai_use_frequency, ai_use_index, ai_chatbot_homework, ai_writing_brainstorm, ai_coding_data_help, ai_study_planning, ai_translate_summarize

Academic AI use refers to how often students use AI tools as part of their coursework, studying, writing, problem solving, coding, planning, or reading. This construct is intentionally broader than the use of one specific tool. In the higher education literature, AI tools include AI assistants, intelligent tutoring systems, learning platforms, automated feedback systems, and other applications that help students complete or organize academic work. This is important for validity because two students may both report frequent AI use while using AI in very different ways.

Measurement is based on a general frequency item and several behavior-specific items. The general item asks how often the student uses AI academically, while the specific items distinguish common use cases such as homework support, writing or brainstorming, coding and data analysis help, study planning, and translating or summarizing readings. These items make it possible to examine whether AI use is best understood as one general exposure or as different types of academic behavior with potentially different implications.

Perceived Benefits of AI for Learning

Variables: ai_benefit_index, benefit_efficiency, benefit_personalized_learning, benefit_feedback, benefit_engagement, benefit_academic_confidence

Perceived AI benefit captures the extent to which students experience AI as helpful for learning. The source studies emphasize that AI may help students by saving time, tailoring explanations, providing immediate feedback, clarifying difficult ideas, structuring information, and increasing engagement. These benefits are especially relevant in college settings because students often balance multiple courses, jobs, commuting, and uneven access to individualized support.

Measurement uses several agreement-style items asking whether AI helps the student finish tasks efficiently, receive explanations that fit how they learn, get useful feedback, feel more engaged, and feel more confident completing academic work. A high score does not necessarily mean AI objectively improves learning; it means the student perceives AI as useful. That distinction matters for validity because perceived usefulness may reflect actual learning support, convenience, confidence, or dependence on the tool.

Concerns About AI Use

Variables: ai_concern_index, concern_overreliance, concern_accuracy, concern_privacy, concern_academic_integrity, concern_reduced_human_interaction

AI concern refers to students' perceived risks or reservations about using AI in education. Vieriu and Petrea report that students' major concerns included inaccurate or imprecise answers, reduced critical thinking, over-dependence on technology, data privacy, and uncertainty about whether AI-generated work truly belongs to the student. These concerns are central to the study because the same technology can be experienced as helpful and unsettling at the same time.

Measurement includes items on over-reliance, accuracy, privacy, academic integrity, and reduced human interaction. The items do not assume that AI is harmful; instead, they measure the degree to which the student worries about specific risks. This is useful for evaluating validity because a high concern score may represent actual negative experiences, greater awareness of ethical issues, unclear course policies, or a tendency to be cautious with new technology.

Critical Thinking, Independent Learning, and Self-Verification

Variables: critical_thinking_index, critical_thinking_confidence, independent_learning_confidence, ai_self_verification

Critical thinking in this dataset refers to students' confidence in evaluating information, learning independently, and checking AI-generated content against other sources. The AI education literature repeatedly frames this as a key issue: AI can quickly produce fluent explanations, but fluent output is not the same as correct, contextualized, or deeply understood output. When students use AI, the educational question is not only whether they receive an answer, but whether they remain actively involved in judging, revising, and understanding that answer.

Measurement focuses on self-reported confidence and verification habits rather than a direct test of reasoning ability. Students report confidence evaluating information independently, confidence learning without copying or relying too heavily on tools, and how consistently they verify AI output. These measures are important for validity because they capture the student's perception and habits, but they may not fully capture actual reasoning performance.

AI Policy Clarity and Permitted Course Use

Variables: ai_policy_clarity, ai_course_use_allowed

AI policy clarity describes whether students feel that their courses or institution clearly explain what counts as acceptable AI use. This matters because the educational meaning of AI use depends on context. Using AI to brainstorm, translate a difficult passage, or check code may be encouraged in one course and restricted in another. Unclear rules can create stress, discourage useful learning support, or make students unsure whether their work is academically honest.

Measurement includes a Likert-scale clarity item and a yes/no item about whether at least one course permits AI use with disclosure. These variables do not measure whether students followed policies perfectly. They measure the student's reported policy environment and perception of rule clarity, which may influence both AI behavior and feelings of uncertainty.

AI Literacy

Variables: ai_literacy_score

AI literacy refers to a student's self-rated ability to use AI tools thoughtfully, understand their limits, and verify output. In the source literature, responsible AI use is often described as requiring training, clear guidelines, and validation practices. A student with stronger AI literacy may be better able to use AI for feedback or learning support while recognizing when the tool may be inaccurate, biased, incomplete, or inappropriate for a task.

Measurement is recorded as a score on an approximately 0 to 100 scale. Because this is self-rated, it should be interpreted as perceived AI literacy rather than an objective skills test. This distinction is important: students who are confident may not always be accurate, and students who are cautious may under-rate their skill.

AI-Related Career Anxiety

Variables: ai_career_anxiety

AI-related career anxiety captures concern that AI will change future job opportunities, make skills obsolete, or increase pressure to keep up with new technology. Although the primary source studies focus on education, career uncertainty is relevant for college students because academic choices are often connected to future employment. Students may view AI not only as a study tool but also as a signal that the labor market is changing.

Measurement uses a 1 to 5 item where higher values indicate more concern about AI and future career prospects. The item should be read as a subjective concern, not as evidence that a student's career path is objectively at risk. It may reflect major, class year, social messages about AI, family expectations, or recent exposure to AI-related news and course content.

Academic Context and Academic Load

Variables: class_year, major_area, units_enrolled, study_hours_week, online_course_ratio, gpa

Academic context variables describe where students are in college and the demands of their coursework. Class year, major area, units enrolled, study hours, online or hybrid course exposure, and self-reported GPA help situate each student's AI use within a broader academic environment. These variables are important for validity because AI use may be shaped by the kinds of assignments students receive and the amount of academic pressure they face.

Measurement combines categorical background information with numeric indicators of workload and academic performance. For example, units enrolled captures formal course load, while study hours captures time spent outside class. Online course ratio records the percentage of current coursework completed online or in hybrid format, which may affect both access to AI tools and the likelihood that students complete assignments digitally.

Economic and Time Pressure Context

Variables: first_gen_college, pell_eligible, financial_stress, employment_hours, commute_minutes

Economic and time-pressure variables describe conditions that may affect both student wellbeing and academic technology use. First-generation college status and Pell eligibility provide broad indicators of educational and economic context. Financial stress, paid work hours, and commute time capture more immediate pressures that can reduce available study time, increase fatigue, or shape why a student might turn to AI for efficiency.

Measurement includes yes/no background indicators, numeric time variables, and a 1 to 5 financial stress item. These variables should be interpreted as context, not as personal deficits. They help students think about construct validity because distress or AI use may reflect broader constraints in a student's life rather than attitudes toward AI alone.

Health Behaviors and Social Support

Variables: sleep_hours, exercise_days, social_support

Sleep, exercise, and social support represent everyday conditions that are closely connected to student wellbeing. They are included because mental health outcomes are not shaped by academic technology alone. A student who sleeps less, works more, feels isolated, or has limited support may report more distress regardless of AI use. Conversely, strong support and healthier routines may help students manage academic demands.

Measurement includes typical nightly sleep hours, days per week with at least 20 minutes of physical activity, and a 1 to 5 perceived social support rating. These measures are brief and self-reported, so they capture broad patterns rather than detailed clinical or behavioral assessments.

Perceived Stress

Variables: perceived_stress_index, stress_overwhelmed, stress_time_pressure, stress_schoolwork, stress_uncertainty

Perceived stress refers to the student's sense of being overwhelmed, under time pressure, strained by assignments or exams, or uncertain about expectations. In the context of AI and education, stress may come from many sources: academic workload, unclear policies, fear of making mistakes, pressure to keep up with new tools, or non-school responsibilities. The construct is not limited to AI-related stress.

Measurement uses four 1 to 5 items covering feeling overwhelmed, time pressure, schoolwork-related stress, and uncertainty about expectations or rules. The composite stress index summarizes these related experiences. Because the items are self-report, they measure perceived stress rather than an external count of stressful events.

Anxiety

Variables: anxiety_index, anxiety_worry, anxiety_restless, anxiety_future, anxiety_trouble_relax

Anxiety in this dataset refers to worry, restlessness, tension, difficulty relaxing, and concerns about future academic or career outcomes. It is framed around students' school and future-related experiences rather than a clinical diagnosis. This is especially relevant in a study of AI because AI can be experienced as both a source of support and a source of uncertainty about academic expectations and future skills.

Measurement uses four 1 to 5 items that ask about worry, restlessness, future-related anxiety, and trouble relaxing. The anxiety index summarizes these items. Higher values indicate more frequent or stronger anxiety-related experiences as reported by the student.

Burnout and Academic Exhaustion

Variables: burnout_index, burnout_exhausted, burnout_cynical, burnout_less_effective, burnout_emotionally_drained

Burnout refers to a pattern of academic exhaustion, emotional drain, detachment from coursework, and reduced feelings of effectiveness as a student. Unlike momentary stress, burnout suggests a more sustained sense that school demands are depleting motivation or capacity. In an AI context, burnout could be related to workload, uncertainty, pressure to constantly adapt, or disappointment when technology does not actually reduce demands.

Measurement uses four 1 to 5 items covering exhaustion, cynicism or detachment, feeling less effective, and feeling emotionally drained. The burnout index averages these items. These items do not diagnose burnout clinically; they summarize the student's reported academic exhaustion and disengagement.

Mental Health Distress Summary

Variables: mental_health_distress_z, high_distress

Mental health distress is a broad summary of negative wellbeing indicators in the dataset. It brings together perceived stress, anxiety, and burnout into a single standardized score. Higher values indicate more overall distress relative to other students in the dataset. This summary is useful for describing overall patterns, but it is less specific than looking separately at stress, anxiety, and burnout.

The binary high distress variable identifies students whose distress summary is relatively elevated. It should not be interpreted as a diagnosis or as evidence that a student needs clinical treatment. It is a descriptive classification based on the survey measures in this dataset.

Positive Wellbeing

Variables: wellbeing_index, wellbeing_optimistic, wellbeing_connected, wellbeing_purpose, wellbeing_calm, wellbeing_belonging

Positive wellbeing captures strengths and positive states such as optimism, connection, purpose, calm, and belonging. Including wellbeing separately from distress is important because mental health is not only the absence of stress, anxiety, or burnout. A student may report stress while also feeling supported and purposeful, or may report low distress but also low connection or belonging.

Measurement uses five 1 to 5 items that ask about optimism, social connection, purpose, calm, and belonging in college. The wellbeing index averages these items, with higher scores indicating more positive wellbeing. As with all self-report measures, the score reflects students' own perceptions at the time of the survey.

Demographic Background Variables

Variables: participant_id, age, gender

Demographic variables provide basic background information. Age and gender may be related to students' educational experiences, technology access, academic responsibilities, or comfort with AI tools. Participant ID is simply an anonymous row identifier used to distinguish records in the dataset.

Measurement includes age in years and self-reported gender categories. These variables should be handled carefully and interpreted as context. Differences across demographic categories can reflect many underlying social and educational processes and should not be over-interpreted without careful attention to validity and confounding.

Variable Dictionary

This table lists each dataset column, its format, the possible values or range, and a student-facing description. Composite variables are averages or standardized summaries of related self-report items; individual item variables retain the original 1 to 5 survey scale.

Variable	Type	Values / Range	Description	Construct group
participant_id	integer	1-400	Anonymous participant identifier. This variable distinguishes rows in the dataset and does not contain personal identity information.	Demographic/background
age	integer	18-38	Student age in years. Age may reflect life stage, academic pathway, work responsibilities, or time since high school.	Demographic/background
gender	factor	woman; man; nonbinary_or_another_identity	Self-reported gender category. Interpret as background context rather than as a biological or causal explanation.	Demographic/background
class_year	ordered factor	first_year < second_year < third_year < fourth_year < fifth_plus	Approximate year in college. This variable captures academic stage and exposure to college-level work.	Academic context
major_area	factor	arts_humanities; business; education_social_science; health_science; stem	Broad field of study. Major area may shape assignment types, AI tool access, and norms for acceptable AI use.	Academic context
first_gen_college	factor	yes/no	Indicates whether the student is the first generation in their family to attend college. This can reflect differences in college navigation support and institutional knowledge.	Economic/educational context
pell_eligible	factor	yes/no	Indicator of Pell Grant eligibility. This is a broad proxy for lower-income financial context and should not be treated as a complete measure of economic status.	Economic/educational context
units_enrolled	integer	6-21	Current academic units enrolled. Higher units generally indicate a heavier formal course load.	Academic context

Variable	Type	Values / Range	Description	Construct group
employment_hours	integer	0-40	Paid work hours per week. Employment can affect available study time, fatigue, financial stress, and motivation for efficiency tools.	Economic/time pressure
study_hours_week	integer	2-35	Hours spent studying outside class per week. This captures time investment, not necessarily study quality.	Academic context
commute_minutes	integer	0-90	One-way commute time in minutes. Commute time can add daily time burden and reduce flexibility.	Economic/time pressure
online_course_ratio	integer	0-100	Percent of current coursework completed online or in hybrid format. Online coursework may increase digital tool use and exposure to AI-enabled platforms.	Academic context
gpa	numeric	2.00-4.00	Self-reported GPA-like academic performance measure. This provides academic context but may include reporting error or grading differences across courses.	Academic context
financial_stress	integer Likert	1-5	Financial stress, where higher values indicate more stress. This captures the student's perceived strain rather than objective income.	Economic/time pressure
sleep_hours	numeric	4.5-9.5	Typical sleep hours per night. Sleep is a general wellbeing context variable and may vary across school and work demands.	Health behavior/support
exercise_days	integer	0-7	Days per week with at least 20 minutes of physical activity. This is a broad health behavior measure.	Health behavior/support
social_support	integer Likert	1-5	Perceived social support, where higher values indicate more support from others.	Health behavior/support
ai_policy_clarity	integer Likert	1-5	Perceived clarity of course or institutional rules for acceptable AI use. Higher values indicate clearer rules.	AI policy context
ai_course_use_allowed	factor	yes/no	Whether the student reports that at least one current course permits AI use with disclosure.	AI policy context

Variable	Type	Values / Range	Description	Construct group
ai_literacy_score	integer	25-95	Self-rated AI literacy and ability to use or verify AI output. Higher values indicate stronger perceived AI literacy.	AI literacy
ai_use_frequency	integer ordinal	1=never; 2=rarely; 3=monthly; 4=weekly; 5=daily	Overall frequency of academic AI use. This variable summarizes how often the student reports using AI for school-related purposes.	AI exposure
ai_career_anxiety	integer Likert	1-5	Concern that AI will reduce future career opportunities or make skills obsolete. Higher values indicate more concern.	AI-related career concern
ai_use_index	numeric mean	1.00-5.00	Average of five academic AI-use items. Higher scores indicate more frequent or broader academic use across multiple tasks.	AI exposure
ai_benefit_index	numeric mean	1.00-5.00	Average of five perceived-benefit items. Higher scores indicate stronger perceived learning benefits from AI.	AI benefit
ai_concern_index	numeric mean	1.00-5.00	Average of five concern/risk items. Higher scores indicate stronger concern about possible negative consequences of AI use.	AI concern
critical_thinking_index	numeric mean	1.00-5.00	Average of critical thinking, independent learning, and self-verification items. Higher scores indicate more confidence and verification-oriented habits.	Critical thinking/self-verification
perceived_stress_index	numeric mean	1.00-5.00	Average of four perceived stress items. Higher scores indicate more school-related stress.	Mental health - stress
anxiety_index	numeric mean	1.00-5.00	Average of four anxiety items. Higher scores indicate more worry, tension, or future-related anxiety.	Mental health - anxiety
burnout_index	numeric mean	1.00-5.00	Average of four burnout/exhaustion items. Higher scores indicate more academic exhaustion or detachment.	Mental health - burnout
mental_health_distress_z	numeric z-score	approximately -2.5 to +2.5	Standardized summary of stress, anxiety, and burnout. Higher values indicate more overall distress relative to other students in the dataset.	Mental health summary

Variable	Type	Values / Range	Description	Construct group
wellbeing_index	numeric mean	1.00-5.00	Average of five positive wellbeing items. Higher scores indicate stronger positive wellbeing.	Positive wellbeing
high_distress	integer binary	0/1	Descriptive indicator for relatively elevated distress in the dataset. This is not a clinical diagnosis.	Mental health summary
ai_chatbot_homework	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequency of using chatbots for homework or problem solving.	AI exposure item
ai_writing_brainstorm	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequency of using AI for writing, brainstorming, or outlining.	AI exposure item
ai_coding_data_help	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequency of using AI for coding, statistics, or data help.	AI exposure item
ai_study_planning	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequency of using AI to plan studying or organize tasks.	AI exposure item
ai_translate_summarize	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequency of using AI to translate, summarize, or simplify readings.	AI exposure item
benefit_efficiency	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Agreement that AI helps the student finish academic tasks more efficiently.	AI benefit item
benefit_personalized_learning	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Agreement that AI helps tailor explanations to the student's way of learning.	AI benefit item
benefit_feedback	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Agreement that AI provides useful immediate feedback.	AI benefit item
benefit_engagement	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Agreement that AI makes coursework feel more engaging.	AI benefit item
benefit_academic_confidence	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Agreement that AI increases confidence in completing academic work.	AI benefit item
concern_overreliance	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Concern about relying too much on AI.	AI concern item
concern_accuracy	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Concern that AI provides inaccurate or misleading information.	AI concern item
concern_privacy	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Concern about privacy or data use when using AI tools.	AI concern item
concern_academic_integrity	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Concern that AI makes it harder to know what counts as honest academic work.	AI concern item
concern_reduced_human_interaction	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Concern that AI reduces interaction with classmates or instructors.	AI concern item
critical_thinking_confidence	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Confidence evaluating information independently.	Critical thinking/self-verification item

Variable	Type	Values / Range	Description	Construct group
independent_learning_confidence	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Confidence learning without copying or relying too heavily on tools.	Critical thinking/self-verification item
ai_self_verification	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	How consistently the student checks AI output against other sources.	Critical thinking/self-verification item
stress_overwhelmed	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling overwhelmed by school responsibilities.	Stress item
stress_time_pressure	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling time pressure from school, work, or other demands.	Stress item
stress_schoolwork	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Stress specifically related to assignments or exams.	Stress item
stress_uncertainty	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Stress from uncertainty about expectations or rules.	Stress item
anxiety_worry	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Frequent worry related to school or the future.	Anxiety item
anxiety_restless	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Restlessness or tension.	Anxiety item
anxiety_future	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Anxiety about future academic or career outcomes.	Anxiety item
anxiety_trouble_relax	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Trouble relaxing because of school or future concerns.	Anxiety item
burnout_exhausted	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Emotional exhaustion from school.	Burnout item
burnout_cynical	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Cynicism or detachment from coursework.	Burnout item
burnout_less_effective	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling less effective as a student.	Burnout item
burnout_emotionally_drained	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling emotionally drained by school demands.	Burnout item
wellbeing_optimistic	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling optimistic about life or school.	Wellbeing item
wellbeing_connected	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling socially connected.	Wellbeing item
wellbeing_purpose	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling a sense of purpose.	Wellbeing item
wellbeing_calm	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling calm and emotionally balanced.	Wellbeing item
wellbeing_belonging	integer Likert	1=strongly disagree/never to 5=strongly agree/very often	Feeling a sense of belonging in college.	Wellbeing item

References Used for Student-Facing Context

- Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: the state of the field. *International Journal of Educational Technology in Higher Education*, 20, Article 22. <https://doi.org/10.1186/s41239-023-00392-8>
- Vieriu, A. M., & Petrea, G. (2025). The impact of artificial intelligence (AI) on students' academic development. *Education Sciences*, 15(3), 343. <https://doi.org/10.3390/educsci15030343>